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Theoretical frameworks and techniques

The study of wave propagation in complex media has long been a central topic in optics, acoustics, and condensed matter physics. In particular, understanding the optical response of ensembles of nanoparticles (NPs) has garnered significant attention due to its relevance in fields such as nanophotonics, plasmonics, and metamaterials. Two main theoretical frameworks are typically employed to model these systems: Multiple Scattering Theories (MSTs) and Effective Medium Theories (EMTs).

The MSTs explicitly accounts for the interactions between individual scatterers and provides an accurate description of wave propagation in structured and disordered media [1–3]. The Foldy-Lax equations, formulated in the mid-20th century, offer a rigorous way to model the self-consistent field scattered by a collection of particles. Another approach to solve the Maxwell’s equation approximately is the formalism of excess current and charge densities, and fields —most commonly found in the context of thermodynamics—, which is suited specifically when the particles are located in an ambient and supported on a substrate.

In the following sections, two MSTs are presented: the Coherent Scattering Model (CSM), based on the Foldy-Lax approach, and the Thin Film Theory, founded on the excess quantities formalism. Both MSTs describe the optical response of disordered systems of spheres, whose centers are coplanar and where the presence of a substrate is considered.

1.1 Coherent Scattering Model

Multiple Scattering Theories (MSTs) provide approximate solutions to Maxwell’s equations when considering the problem of light scattered by an ensemble of particles illuminated by a monochromatic plane wave [1–3]. In particular, the Coherent Scattering Model (CSM) is a MST that provides an expression for the amplitude coefficients of reflection and transmission of light scattered in the coherent direction for a disordered ensemble of spherical particles, whose centers are coplanar [3–5].

Within the MST framework, the optical response of a disordered array of arbitrary particles, illuminated by an incident monochromatic plane wave $\mathbf{E}^{\text{inc}}(\mathbf{r}, \omega)$ with angular frequency ω and propagation direction $\hat{\mathbf{k}}^{\text{inc}}$, is determined by the excitation field $\mathbf{E}^{\text{exc}}(\mathbf{r}, \omega)$ acting on the

k -th particle in a collection of N particles. This field is equal to the sum of the incident electric field and the induced electric field $\mathbf{E}_\ell^{\text{ind}}(\mathbf{r}, \omega)$ —which includes both the field inside the particle and the scattered field outside of it— due to the other elements of the ensemble but not itself [4, 5], that is

$$\mathbf{E}_k^{\text{exc}}(\mathbf{r}) = \mathbf{E}^{\text{inc}}(\mathbf{r}) + \sum_{\ell \neq k}^N \mathbf{E}_\ell^{\text{ind}}(\mathbf{r}), \quad (1.1)$$

where the harmonic dependence is intrinsically implied. Since all particles are excited by the same interaction, the electric field induced in the ℓ -th particle is given by [4, 5]

$$\mathbf{E}_\ell^{\text{ind}}(\mathbf{r}) = \int d^3 r' \mathbb{G}(\mathbf{r}, \mathbf{r}') \int d^3 r'' \mathbb{T}(\mathbf{r}' - \mathbf{r}_\ell, \mathbf{r}'' - \mathbf{r}_\ell) \mathbf{E}_\ell^{\text{exc}}(\mathbf{r}''), \quad (1.2)$$

where $\mathbf{r}_\ell$ is the position of the center of the ℓ -th particle, $\mathbb{G}(\mathbf{r}, \mathbf{r}')$ is the dyadic Green function —solution to the vectorial Helmholtz equation with the product of the unitary dyadic and a Dirac delta centered at $\mathbf{r}'$ as a source [6]— and $\mathbb{T}$ is the transition operator, or matrix T , that linearly relates the scattered electric field of a particle to the electric field that excites it [6]. The integrals in Eq. (1.2) are performed over a volume V , described by the positions $\mathbf{r}'$ and $\mathbf{r}''$, such that the center $\mathbf{r}_\ell$ of the ℓ -th particle is located within V [4].

The total electric field $\mathbf{E}(\mathbf{r})$ is equal to the sum of $\mathbf{E}^{\text{inc}}(\mathbf{r})$ and the N contributions of the induced electric field $\mathbf{E}_\ell^{\text{ind}}(\mathbf{r})$ for each particle, which is determined by solving the system of N equations given by Ecs. (1.1) and (1.2). For an ensemble of randomly located particles, the configurational average of the total electric field $\langle \mathbf{E}(\mathbf{r}) \rangle$ is calculated by considering the probability of occurrence of each combination in which the centers $\mathbf{r}_\ell$ of the elements can be located [4, 5]:

$$\langle \mathbf{E}(\mathbf{r}) \rangle = \mathbf{E}^{\text{inc}}(\mathbf{r}) + \sum_{\ell=1}^N \langle \mathbf{E}_\ell^{\text{ind}}(\mathbf{r}) \rangle = \mathbf{E}^{\text{inc}}(\mathbf{r}) + \sum_{\ell=1}^N \left(\prod_{k=1}^N \int d^3 r_k \rho(\mathbf{R}) \mathbf{E}_\ell^{\text{ind}}(\mathbf{r}) \right), \quad (1.3)$$

with $\rho(\mathbf{R})$ is the probability density that the ensemble is in a specific spatial configuration given by $\mathbf{R} = (\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N)^T$. Configurationally averaging Eq. (1.2) gives the average contribution of the induced electric field by the ℓ -th particle, which can be written as [4, 5]

$$\langle \mathbf{E}_\ell^{\text{ind}}(\mathbf{r}) \rangle = \int d^3 r' \mathbb{G}(\mathbf{r}, \mathbf{r}') \int d^3 r'' \int d^3 r_\ell \rho(\mathbf{r}_\ell) \mathbb{T}(\mathbf{r}' - \mathbf{r}_\ell, \mathbf{r}'' - \mathbf{r}_\ell) \langle \mathbf{E}_\ell^{\text{exc}}(\mathbf{r}'', \mathbf{R}) \rangle_\ell, \quad (1.4a)$$

$$\langle \mathbf{E}_\ell^{\text{exc}}(\mathbf{r}'', \mathbf{R}) \rangle_\ell = \prod_{\substack{k=1 \\ k \neq \ell}}^N \int d^3 r_k \rho(\mathbf{R}|\mathbf{r}_\ell) \mathbf{E}_\ell^{\text{exc}}(\mathbf{r}''), \quad (1.4b)$$

where $\rho(\mathbf{r}_\ell)$ is the probability density of finding the center of the ℓ -th particle within the volume V , and Eq. (1.4b) represents the configurational average of the excitation electric field $\mathbf{E}_\ell^{\text{exc}}$ for the ℓ -th particle, considering that its position is fixed in V ; the conditional probability density $\rho(\mathbf{R}|\mathbf{r}_\ell)$ describes such constriction. This expression can be further analyzed by substituting Eq. (1.1) into Eq. (1.4b) and following an analogous procedure to that of Eqs. (1.4) with the

Fig. 1.1: Diagram of the particle ensemble considered in the CSM, consisting of a collection of N identical spherical particles of radius a whose centers are randomly located within the integration volume V : an infinite film centered at the plane $z = 0$ and of thickness d , which tends to zero to reproduce the case of a two-dimensional array. The system, embedded in a dielectric matrix, is illuminated by a monochromatic plane wave incident on the film at an angle θ .

index exchanges $k \rightarrow \ell$ and $\ell \rightarrow m$, yielding

$$\langle \mathbf{E}_\ell^{\text{exc}}(\mathbf{r}'', \mathbf{R}) \rangle_\ell = \mathbf{E}^{\text{inc}}(\mathbf{r}'') + \sum_{\substack{m=1 \\ m \neq \ell}}^N \int d^3 r' \mathbb{G}(\mathbf{r}', \mathbf{r}'') \times \int d^3 r''' \int d^3 r_m \rho(\mathbf{r}_m) \mathbb{T}(\mathbf{r}' - \mathbf{r}_m, \mathbf{r}''' - \mathbf{r}_m) \langle \mathbf{E}_m^{\text{exc}}(\mathbf{r}''', \mathbf{R}) \rangle_{\ell, m}, \quad (1.5a)$$

$$\langle \mathbf{E}_m^{\text{exc}}(\mathbf{r}'', \mathbf{R}) \rangle_{\ell, m} = \prod_{\substack{n=1 \\ n \neq \ell, m}}^N \int d^3 r_n \rho(\mathbf{R} | \mathbf{r}_\ell, \mathbf{r}_m) \mathbf{E}_n^{\text{exc}}(\mathbf{r}''), \quad (1.5b)$$

where Eq. (1.5b) represents the configurational average of the excitation field $\mathbf{E}_m^{\text{exc}}(\mathbf{r}'')$ acting on the m -th particle, considering the conditional probability density $\rho(\mathbf{R} | \mathbf{r}_\ell, \mathbf{r}_m)$ that the system is in a configuration $\mathbf{R}$ with the positions $\mathbf{r}_\ell$ and $\mathbf{r}_m$ fixed in space [4, 5]. By continuing this process for all N spheres in the ensemble, a set of N equations known as the Foldy-Lax hierarchy [1] can be constructed. Solving this hierarchy leads to the solution of the averaged electric field at a point $\mathbf{r}$, taking multiple scattering effects into account. The hierarchical equations can be truncated at different orders, reducing them to invertible systems of equations by applying approximations to how multiple scattering excites the elements of the ensemble.

To truncate the Foldy-Lax hierarchy and determine the total averaged electric field, the CSM imposes two approximations at different orders. First, the Single Scattering Approximation (SSA) is considered, reducing the Foldy-Lax hierarchy to a single equation to be solved by neglecting multiple scattering effects [4]. Specifically, in the SSA, $\langle \mathbf{E}_\ell^{\text{exc}}(\mathbf{r}'', \mathbf{R}) \rangle_\ell$ is forced to equal the incident electric field $\mathbf{E}^{\text{inc}}(\mathbf{r})$ in Eq. (1.4b) [1, 5]. The second approximation in the CSM is the Quasi-Crystalline Approximation (QSA), where the Foldy-Lax hierarchy is rewritten as a system of two equations by assuming that the configurational average for cases with one and two fixed particles is approximately equal, meaning that Eqs. (1.4b) and (1.5) are set equal [1, 4]. While the SSA is suitable for dilute disordered arrays, as it disregards multiple scattering, the QSA adapts to denser ensembles—up to a coverage fraction of 60% [1]—because, under this condition, the average with two fixed particles does not significantly differ from the average with one fixed particle due to the presence of the remaining particles [4].

Once the approximation orders have been determined to truncate the Foldy-Lax hierarchy, the CSM introduces a series of assumptions about the particle ensemble to solve the resulting system of equations for each approximation. In particular, the CSM considers that the particle ensemble consists of $N \gg 1$ identical spherical particles of radius a —embedded in a dielectric medium known as the matrix—whose centers $\mathbf{r}_\ell$ are within the integration volume V , which consists of an infinite film of thickness d centered at $z = 0$, with a uniform probability density $\rho(\mathbf{r}_\ell) = 1/V$, and assumes that the volume V_ℓ of each sphere is negligible compared to V [4]; a schematic of the system is shown in Fig. 1.1. Under these assumptions, the electric field induced

by the ensemble particles in the SSA can be computed by substituting in Eq. (1.4) the dyadic Green function $\mathbb{G}$ with its plane wave representation and performing the configurational average, along with the limit $d \rightarrow 0$ to consider the case of a two-dimensional array [4]. This procedure results in

$$\langle \mathbf{E}_{\text{SSA}}^{\text{ind}}(\mathbf{r}) \rangle = \begin{cases} -\alpha S_j(\pi - 2\theta) \exp(i\mathbf{k}_{\text{coh}}^{\text{r}} \cdot \mathbf{r}) \mathbf{E}^{\text{inc}}(\mathbf{r}), & z > 0, \\ -\alpha S(0) \exp(i\mathbf{k}_{\text{coh}}^{\text{t}} \cdot \mathbf{r}) \mathbf{E}^{\text{inc}}(\mathbf{r}), & z < 0, \end{cases} \quad \text{with} \quad \alpha = \frac{2\Theta}{x^2 \cos \theta}, \quad (1.6)$$

where θ is the incidence angle at which the incident plane wave illuminates the two-dimensional array, $S_j(\vartheta)$ are the elements of the Mie scattering matrix [7]—which depend on the refractive index of the matrix, the sphere, and its size—evaluated at ϑ , with $j = 1$ for s -polarization and $j = 2$ for p -polarization. The term Θ corresponds to the coverage fraction of the array, and $x = k^{\text{ind}} a$ is the size parameter with k^{ind} being the magnitude of the incident field wave vector propagating in the matrix [4, 5]. The elements $S_j(\vartheta)$ are evaluated at $\vartheta = \pi - 2\theta$ and $\vartheta = 0$ as they correspond to the coherent scattering directions given by the law of reflection and Snell's law, respectively. From Eq. (1.6), it is shown that under the SSA, the two-dimensional array of spherical particles scatters light primarily in coherent directions, as the diffuse components interfere destructively [4]. Consequently, reflection $r_{\text{SSA}}^{(j)}$ and transmission $t_{\text{SSA}}^{(j)}$ amplitude coefficients can be defined for the two-dimensional array similarly to the Fresnel coefficients, given by

$$r_{\text{SSA}}^{(j)}(\theta) = -\alpha S_j(\pi - 2\theta), \quad \text{and} \quad t_{\text{SSA}}^{(j)}(\theta) = 1 - \alpha S(0). \quad (1.7)$$

In the case of the QSA, the system of equations to be solved consists of Eqs. (1.4) and (1.5), considering that Eqs. (1.4b) and (1.5b) are equal. To solve this system of equations, the ensemble is assumed to have the same characteristics as in the SSA case and additionally that the surroundings of each sphere are identical on average [5]. Therefore, the system of equations can be solved by proposing the *ansatz*

$$\langle \mathbf{E}_{\ell}^{\text{exc}}(\mathbf{r}) \rangle_{\ell} = \mathbf{E}_{\text{r}}^{\text{exc}}(\mathbf{r}) \exp(i\mathbf{k}_{\text{coh}}^{\text{r}} \cdot \mathbf{r}) + \mathbf{E}_{\text{t}}^{\text{exc}}(\mathbf{r}) \exp(i\mathbf{k}_{\text{coh}}^{\text{t}} \cdot \mathbf{r}) \quad (1.8)$$

that is, in the CSM, it is assumed that $\langle \mathbf{E}_{\ell}^{\text{exc}}(\mathbf{r}) \rangle_{\ell}$ consists of a contribution from an electric field propagating in the coherent reflection direction, denoted as $\mathbf{E}_{\text{r}}^{\text{exc}}$, and another in the transmission direction, denoted by $\mathbf{E}_{\text{t}}^{\text{exc}}$ [5]. These coherent contributions behave according to the SSA [Eq. (1.6)], so each will have a reflected and transmitted component. The sum of these contributions determines the induced electric field under the QSA. Solving the system of equations and substituting in Eq. (1.9), the amplitude coefficients of reflection $r_{\text{CSM}}^{(j)}$ and transmission $t_{\text{CSM}}^{(j)}$ can be defined, considering multiple scattering [4, 5]. These coherent contributions behave according to the SSA [Eq. (1.6)], so each of them will have a reflected and a transmitted component. The sum of these contributions determines the induced electric field under the QSA, whose expression is

$$\langle \mathbf{E}_{\text{QSA}}^{\text{ind}}(\mathbf{r}) \rangle = \begin{cases} -\alpha \left(S_j(0) \|\mathbf{E}_{\text{r}}^{\text{exc}}(\mathbf{r})\| + S(\pi - 2\theta) \|\mathbf{E}_{\text{t}}^{\text{exc}}(\mathbf{r})\| \right) \exp(i\mathbf{k}_{\text{coh}}^{\text{r}} \cdot \mathbf{r}) \hat{\mathbf{E}}^{\text{inc}}, & z > 0, \\ -\alpha \left(S_j(\pi - 2\theta) \|\mathbf{E}_{\text{r}}^{\text{exc}}(\mathbf{r})\| + S(0) \|\mathbf{E}_{\text{t}}^{\text{exc}}(\mathbf{r})\| \right) \exp(i\mathbf{k}_{\text{coh}}^{\text{t}} \cdot \mathbf{r}) \hat{\mathbf{E}}^{\text{inc}}, & z < 0, \end{cases} \quad (1.9)$$

where $\hat{\mathbf{E}}^{\text{inc}}$ indicates the polarization of the incident electric field. Since in the QSA the expressions of Eq. (1.5) are equal for the electric field exciting a particle, then the contributions of the induced field in the coherent direction satisfy [5]

$$\mathbf{E}_r^{\text{exc}}(\mathbf{r}) = -\frac{\alpha}{2} \left(S_j(0) \|\mathbf{E}_r^{\text{exc}}(\mathbf{r})\| + S(\pi - 2\theta) \|\mathbf{E}_t^{\text{exc}}(\mathbf{r})\| \right) \hat{\mathbf{E}}^{\text{inc}}, \quad (1.10a)$$

$$\mathbf{E}_t^{\text{exc}}(\mathbf{r}) = \mathbf{E}^{\text{inc}}(\mathbf{r}) - \frac{\alpha}{2} \left(S_j(\pi - 2\theta) \|\mathbf{E}_r^{\text{exc}}\| + S(0) \|\mathbf{E}_t^{\text{exc}}(\mathbf{r})\| \right) \hat{\mathbf{E}}^{\text{inc}}. \quad (1.10b)$$

Solving the system of equations for $\mathbf{E}_r^{\text{exc}}(\mathbf{r})$ and $\mathbf{E}_t^{\text{exc}}(\mathbf{r})$ shown in Eq. (1.10), and substituting them in Eq. (1.9), expressions for the reflection $r_{\text{CSM}}^{(j)}$ and transmission $t_{\text{CSM}}^{(j)}$ amplitude coefficients are defined, considering multiple scattering. These expressions are [4, 5]

$$r_{\text{CSM}}^{(j)}(\theta) = \frac{-\alpha S_j(\pi - 2\theta)}{1 + \alpha S(0) + \frac{1}{4}\alpha^2 [S^2(0) - S_j^2(\pi - 2\theta)]}, \quad (1.11a)$$

$$t_{\text{CSM}}^{(j)}(\theta) = \frac{1 - \frac{1}{4}\alpha^2 [S^2(0) - S_j^2(\pi - 2\theta)]}{1 + \alpha S(0) + \frac{1}{4}\alpha^2 [S^2(0) - S_j^2(\pi - 2\theta)]}, \quad (1.11b)$$

where $j = 1$ for s -polarization and $j = 2$ for p -polarization. The assumption that spherical particles are identical, imposed in both the SSA and QSA, can be relaxed if the radius a of the particles follows a size distribution density $\rho(a)$ and the field exciting each sphere is identical for all, except for a phase difference of $\pm 2ik_z^{\text{inc}}a$ caused by size variation [8]. In this case, the average over the particle radius a is computed in Eq. (1.10), incorporating the phase difference by multiplying the right-hand side of Eq. (1.10) by $\exp(\pm 2ik_z^{\text{inc}}a)$ —with $k_z^{\text{ind}} = k^{\text{ind}} \cos \theta$ —, where the positive sign corresponds to the reflected contribution and the negative to the transmitted one. Following the analogous procedure described earlier, the reflection and transmission amplitude coefficients considering size polydispersity in the two-dimensional array are given by [8].

$$r_{\text{CSM}}^{(j)}(\theta) = \frac{-\beta_+^{(j)}(\theta)}{1 + \beta_F + \frac{1}{4} [\beta_F^2 - \beta_+^{(j)}(\theta) \beta_-^{(j)}(\theta)]}, \quad (1.12a)$$

$$t_{\text{CSM}}^{(j)}(\theta) = \frac{1 - \frac{1}{4} [\beta_F^2 - \beta_+^{(j)}(\theta) \beta_-^{(j)}(\theta)]}{1 + \beta_F + \frac{1}{4} [\beta_F^2 - \beta_+^{(j)}(\theta) \beta_-^{(j)}(\theta)]}, \quad (1.12b)$$

where

$$\beta_F(\theta) = \eta \int_0^\infty \rho(a) S(0) da \quad \text{and} \quad \beta_\pm^{(j)}(\theta) = \eta \int_0^\infty \rho(a) S_j(\pi - 2\theta) \exp(\pm 2ik_z^{\text{inc}}a) da, \quad (1.13)$$

with $\eta = 2\Theta/(x_0^2 \cos \theta)$ and where $x_0 = k^{\text{ind}} a_0$ is the size parameter of a sphere with radius a_0 , which is the most probable value according to the size distribution density $\rho(a)$.

The expressions in Eq. (1.12) allow describing the response of a disordered metasurface of spherical NPs under conditions close to experimental ones by introducing the presence of a substrate. To achieve this, multiple reflections between the substrate-matrix interface and the matrix-metasurface interface are considered, assuming that the latter responds according to Eq. (1.12) when separated from the substrate by a distance $\Delta \rightarrow 0$ [8]. In particular, to compare

the theoretical response with reflectance measurements in an internal incidence scheme, the amplitude reflection coefficient of the entire system (metasurface with substrate) is given by [8]

$$r_{\text{CSM}}^{(j)}(\theta_i) = \frac{r_{\text{sm}}^{(j)}(\theta_i) + r_{\text{CSM}}^{(j)}(\theta_t)}{1 + r_{\text{sm}}^{(j)}(\theta_i)r_{\text{CSM}}^{(j)}(\theta_t)}, \quad (1.14)$$

where r_{sm} is the Fresnel reflection coefficient between the substrate and the matrix for s (p) polarization if $j = 1$ ($j = 2$), and where θ_i is the angle of incidence of the plane wave illuminating the interface between the substrate and the matrix, and θ_t is the transmission angle upon crossing this interface and thus the angle at which it illuminates the metasurface.

1.2 Thin Film Theory

The Thin Film Theory (TFT) is theoretical framework used to describe the optical properties of thin films and small particles deposited on a substrate. The TFT is a MST that returns an approximated solution to Maxwells equation similarly to Fresnel's equations where two semi-infinite media enclose a region of the space, herby identified as a thin film, with a previously unspecified composition. To describe the optical properties of the thin film, the excess field formalism is employed thus introducing a polarization and magnetization fields of the thin the film, which can model an ensemble of metallic nano-particles through constitutive equations. This approach modifies Maxwell's equation at the interfaces of the system, thus determining explicitly the discontinuities of the electromagnetic field at the boundaries of the thin film.

To study the reflection and transmission of an electromagnetic plane wave illuminating an arbitrary thin film of thickness d centered at $z = 0$ in between two media characterized by their bulk dielectric functions $\varepsilon^{\geq} = (n^{\geq})^2$, it is proposed that the total electric field $\mathbf{E}(\mathbf{r})$ is decomposed in three contributions: inside the thin film and in the media above and below it denoted by the symbols ($>$) and ($<$), respectively. Such decomposition is written as follows

$$\mathbf{E}(\mathbf{r}) = \mathbf{E}^>(\mathbf{r})\Theta(z) + \mathbf{E}^<(\mathbf{r})\Theta(-z) + \mathbf{E}^{\text{excess}}(\mathbf{r}), \quad (1.15)$$

where the harmonic dependency in the angular frequency ω is implicitly included in the employed notation and $\Theta(z)$ is the Heaviside step function; the electric fields $\mathbf{E}^{\geq}$ are considered to be radiation fields. The term $\mathbf{E}^{\text{excess}}(\mathbf{r})$ is the excess electric field and its contribution is non-zero in the neighborhood of the thin film since $\mathbf{E}(\mathbf{r}) \rightarrow \mathbf{E}^{\geq}(\mathbf{r})$ when $z \rightarrow \pm\infty$. Thus, the whole effect of the thin film in the electromagnetic properties of the system can be described by considering the total excess electric field in the film $\mathbf{E}^{\text{film}}(\mathbf{r}, z = 0)$ in the limit when $d \rightarrow 0$, thus, yielding the expression

$$\mathbf{E}(\mathbf{r}) = \mathbf{E}^>(\mathbf{r})\Theta(z) + \mathbf{E}^<(\mathbf{r})\Theta(-z) + \mathbf{E}^{\text{film}}(\mathbf{r}_{\parallel})\delta(z), \quad (1.16)$$

with $\delta(z)$ the Dirac delta function and

$$\mathbf{E}^{\text{film}}(\mathbf{r}_{\parallel}) = \int_{-\infty}^{\infty} \mathbf{E}^{\text{excess}}(\mathbf{r}) dz = \left(\int_{-\infty}^{\infty} \mathbf{E}^{\text{excess}}(\mathbf{r}) \exp(ik_z z) dz \right) \Big|_{k_z=0}, \quad (1.17)$$

where $\mathbf{r}_{\parallel}$ is a vector evaluated at the plane $z = 0$, and where it is explicitly shown that $\mathbf{E}^{\text{film}}(\mathbf{r}_{\parallel})$ equals the Fourier transform of the excess electric field in the z variable evaluated at $k_z = 0$. The same decomposition employed in Eqs. (1.15)–(1.17) for $\mathbf{E}(\mathbf{r})$ applies to the magnetic field $\mathbf{B}(\mathbf{r})$, as well as any other scalar or vector field relevant to the system, for example, the electric displacement field $\mathbf{D}(\mathbf{r})$, the $\mathbf{H}(\mathbf{r})$ field, and the external charge $\rho_{\text{ext}}(\mathbf{r})$ and current $\mathbf{J}_{\text{ext}}(\mathbf{r})$ densities. By substituting such definitions of the fields into Maxwell's equations, it follows that the excess quantities obey the following expressions

$$\nabla \cdot \mathbf{D}^{\text{excess}}(\mathbf{r}) + [D_{\perp}^>(\mathbf{r}_{\parallel}) - D_{\perp}^<(\mathbf{r}_{\parallel})]\delta(z) = \rho_{\text{ext}}^{\text{excess}}(\mathbf{r}), \quad (1.18a)$$

$$\nabla \cdot \mathbf{B}^{\text{excess}}(\mathbf{r}) + [B_{\perp}^>(\mathbf{r}_{\parallel}) - B_{\perp}^<(\mathbf{r}_{\parallel})]\delta(z) = 0, \quad (1.18b)$$

$$\nabla \times \mathbf{E}^{\text{excess}}(\mathbf{r}) + \hat{\mathbf{e}}_{\perp} \times [\mathbf{E}_{\parallel}^>(\mathbf{r}_{\parallel}) - \mathbf{E}_{\parallel}^<(\mathbf{r}_{\parallel})]\delta(z) = +i\omega\mathbf{B}^{\text{excess}}(\mathbf{r}), \quad (1.18c)$$

$$\nabla \times \mathbf{H}^{\text{excess}}(\mathbf{r}) + \hat{\mathbf{e}}_{\perp} \times [\mathbf{H}_{\parallel}^>(\mathbf{r}_{\parallel}) - \mathbf{H}_{\parallel}^<(\mathbf{r}_{\parallel})]\delta(z) = \mathbf{J}_{\text{ext}}^{\text{excess}}(\mathbf{r}) - i\omega\mathbf{D}^{\text{excess}}(\mathbf{r}), \quad (1.18d)$$

which are obtained by considering that the fields in the semi-infinite media are solution to Maxwell's equations each and that the step and delta functions are related by the relation $d\Theta(\pm z)/dz = \pm\delta(z)$. The subscript ($\parallel$) correspond to the parallel components of the fields relative to the thin film, ($\perp$) to their normal components to the film; $\hat{\mathbf{e}}_{\perp}$ is the unitary vector in this direction. The terms in between brackets in Eqs. (1.18) are the boundary conditions of the electromagnetic fields evaluated at the thin film and they are modified from the known case of a sharp interface—which yield Fresnel's equations—by the excess quantities.

The boundary conditions of the electromagnetic field in the TFT formalism can be determined by calculating the Fourier transform of Eqs. (1.18) in the z variable and then considering the particular case of $k_z = 0$ so that all excess quantities are interchanged for their total contribution according to Eq. (1.17). Additionally, total excess polarization $\mathbf{P}^{\text{film}}(\mathbf{r}_{\parallel})$ and magnetization $\mathbf{M}^{\text{film}}(\mathbf{r}_{\parallel})$ are defined in terms of the total excess electromagnetic fields, thus the boundary conditions in the absence of external sources — $\mathbf{J}_{\text{ext}}(\mathbf{r}) = \mathbf{0}$ and $\rho_{\text{ext}}(\mathbf{r}) = 0$ — can be written as

$$[D_{\perp}^>(\mathbf{r}_{\parallel}) - D_{\perp}^<(\mathbf{r}_{\parallel})] = -\nabla_{\parallel} \cdot \mathbf{P}_{\parallel}^{\text{film}}(\mathbf{r}_{\parallel}), \quad (1.19a)$$

$$[B_{\perp}^>(\mathbf{r}_{\parallel}) - B_{\perp}^<(\mathbf{r}_{\parallel})] = -\nabla_{\parallel} \cdot \mathbf{M}_{\parallel}^{\text{film}}(\mathbf{r}_{\parallel}), \quad (1.19b)$$

$$[\mathbf{E}_{\parallel}^>(\mathbf{r}_{\parallel}) - \mathbf{E}_{\parallel}^<(\mathbf{r}_{\parallel})] = -i\omega\hat{\mathbf{e}}_{\perp} \times \mathbf{M}_{\parallel}^{\text{film}}(\mathbf{r}_{\parallel}) - \nabla_{\parallel} P_{\perp}^{\text{film}}(\mathbf{r}_{\parallel}), \quad (1.19c)$$

$$[\mathbf{H}_{\parallel}^>(\mathbf{r}_{\parallel}) - \mathbf{H}_{\parallel}^<(\mathbf{r}_{\parallel})] = +i\omega\hat{\mathbf{e}}_{\perp} \times \mathbf{P}_{\parallel}^{\text{film}}(\mathbf{r}_{\parallel}) - \nabla_{\parallel} M_{\perp}^{\text{film}}(\mathbf{r}_{\parallel}), \quad (1.19d)$$

where $\nabla_{\parallel}$ and $\nabla_{\parallel} \cdot$ are the bidimensional gradient and divergence operators on the thin film, respectively, and

$$\mathbf{P}^{\text{film}}(\mathbf{r}_{\parallel}) = \mathbf{D}_{\parallel}^{\text{film}}(\mathbf{r}_{\parallel}) - \varepsilon_0 E_{\perp}^{\text{film}}(\mathbf{r}_{\parallel})\hat{\mathbf{e}}_{\perp} \quad (1.20)$$

$$\mathbf{M}^{\text{film}}(\mathbf{r}_{\parallel}) = \frac{1}{\mu_0} \mathbf{B}_{\parallel}^{\text{film}}(\mathbf{r}_{\parallel}) - H_{\perp}^{\text{film}}(\mathbf{r}_{\parallel})\hat{\mathbf{e}}_{\perp} \quad (1.21)$$

with ε_0 (μ_0) the electric permittivity (magnetic permeability) of the vacuum. It is noteworthy that the definition of $\mathbf{P}^{\text{film}}(\mathbf{r}_{\parallel})$ in Eq. (1.20) is not just a direct identification of the procedure from Eqs. (1.18) to Eqs. (1.19) but it is based on the non-continuity of $\mathbf{D}_{\parallel}(\mathbf{r})$ and $E_{\perp}(\mathbf{r})$ across

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boundaries, which give rise to an excess polarization on the film unlike $D_{\perp}(\mathbf{r})$ and $\mathbf{E}_{\parallel}(\mathbf{r})$, whose continuity do not have such an effect on the system. An analogous argument is performed on the definition of the magnetization field in Eq. (1.21).

The boundary conditions of the electromagnetic fields shown in Eq. (1.19) reproduce Fresnel's equations, where the film corresponds to a planar sharp interface between two linear homogeneous media, when $\mathbf{P}^{\text{film}}(\mathbf{r}_{\parallel}) = \mathbf{M}^{\text{film}}(\mathbf{r}_{\parallel}) = \mathbf{0}$. To introduce a thin film with a known structure, the total excess polarization and magnetization [Eqs. (1.20) and (1.21)] can be described by constitutive equations as an expansion to different orders of the electromagnetic fields. The particular case where the thin film corresponds to a collection of identical plasmonic scatters, with a size where no magnetic response is expected, the general constitutive relations to Eqs. (1.20) and (1.21) can be express to a linear order as

$$\mathbf{P}^{\text{film}}(\mathbf{r}_{\parallel}) = \gamma \langle \mathbf{E}_{\parallel}^{\text{film}}(\mathbf{r}_{\parallel}) \rangle + \beta \langle D_{\perp}^{\text{film}}(\mathbf{r}_{\parallel}) \rangle \hat{\mathbf{e}}_{\perp} \quad \text{and} \quad \mathbf{M}^{\text{film}}(\mathbf{r}_{\parallel}) = \mathbf{0} \quad (1.22)$$

where γ and β are known as the film susceptibilities, which encodes the physical properties of the ensemble of scatterers, and where the operator $\langle \cdot \rangle$ is the average on the film given as $\langle a^{\text{film}}(\mathbf{r}_{\parallel}) \rangle = (1/2)[a^{>}(\mathbf{r}_{\parallel}) + a^{<}(\mathbf{r}_{\parallel})]$, for either a scalar or vector quantity $a^{\lessgtr}(\mathbf{r})$.

The TFT employs the excess field formalism to determine the optical response of a metasurface embedded in a matrix and supported on a substrate, as shown in Fig. ??, both non-absorbing nor magnetic ($\mu^{\lessgtr} = \mu_0$), by calculating amplitude reflection r_{TFT} and transmission t_{TFT} coefficients from Eqs. (1.19) and (1.22), considering that the thin film consists in a disordered ensemble of identical plasmonic spherical NPs. In an schema of internal illumination, the electric field below the thin film is the superposition of an incident and a reflected electromagnetic plane waves $\mathbf{E}^{\text{inc}}(\mathbf{r})$ and $\mathbf{E}^{\text{r}}(\mathbf{r})$ traveling in the $\mathbf{k}^{\text{inc}}$ and $\mathbf{k}^{\text{r}}$, respectively, within the substrate characterized by the dielectric function $\varepsilon^{<} = \varepsilon_i = n_i^2$, while above the electric field is given by the plane wave $\mathbf{E}^{\text{t}}(\mathbf{r})$ traveling in the $\mathbf{k}^{\text{t}}$ direction within the matrix with $\varepsilon^{>} = \varepsilon_t = n_t^2$. To determine the amplitude coefficients under the TFT the cases of an s and a p polarized incident electromagnetic wave are treated separately.

In general, when an electromagnetic plane wave with an angular frequency ω travels from a substrate to a thin film in the direction $\mathbf{k}^{\text{inc}}$ at an angle θ_i , relative to the normal direction to the thin film, the reflected (transmitted) plane wave does it in the direction of $\mathbf{k}^{\text{r}}$ ($\mathbf{k}^{\text{t}}$) at the angle θ_r (θ_t). Thus, the wave vectors of the electromagnetic plane waves are

$$\mathbf{k}^{\text{inc}} = n_i \frac{\omega}{c} (\sin \theta_i \hat{\mathbf{e}}_x + \cos \theta_i \hat{\mathbf{e}}_{\perp}), \quad (1.23a)$$

$$\mathbf{k}^{\text{r}} = n_i \frac{\omega}{c} (\sin \theta_r \hat{\mathbf{e}}_x - \cos \theta_r \hat{\mathbf{e}}_{\perp}), \quad (1.23b)$$

$$\mathbf{k}^{\text{t}} = n_t \frac{\omega}{c} (\sin \theta_t \hat{\mathbf{e}}_x + \cos \theta_t \hat{\mathbf{e}}_{\perp}), \quad (1.23c)$$

where $\hat{\mathbf{e}}_{\perp} = \hat{\mathbf{e}}_z$. For an s polarized electromagnetic plane wave [see Fig. ??], the electric field above and below the thin film is

$$\mathbf{E}^{<}(\mathbf{r}) = \left[E^{\text{inc}} \exp(i\mathbf{k}^{\text{inc}} \cdot \mathbf{r}) + E_{\text{r}}^{(s)} \exp(i\mathbf{k}^{\text{r}} \cdot \mathbf{r}) \right] \hat{\mathbf{e}}_y \quad (1.24a)$$

$$\mathbf{E}^{>}(\mathbf{r}) = E_{\text{t}}^{(s)} \exp(i\mathbf{k}^{\text{t}} \cdot \mathbf{r}) \hat{\mathbf{e}}_y \quad (1.24b)$$

while for a p polarized plane wave, the electric field is given by

$$\begin{aligned} \mathbf{E}^{<}(\mathbf{r}) = & \left[-E^{\text{inc}} \cos \theta_i \exp(i\mathbf{k}^{\text{inc}} \cdot \mathbf{r}) + E_r^{(p)} \cos \theta_r \exp(i\mathbf{k}^r \cdot \mathbf{r}) \right] \hat{\mathbf{e}}_x + \\ & + \left[E^{\text{inc}} \sin \theta_i \exp(i\mathbf{k}^{\text{inc}} \cdot \mathbf{r}) + E_r^{(p)} \sin \theta_r \exp(i\mathbf{k}^r \cdot \mathbf{r}) \right] \hat{\mathbf{e}}_{\perp} \end{aligned} \quad (1.25a)$$

$$\mathbf{E}^{>}(\mathbf{r}) = \left[-E_t^{(p)} \cos \theta_t \exp(i\mathbf{k}^t \cdot \mathbf{r}) \right] \hat{\mathbf{e}}_x + \left[-E_t^{(p)} \sin \theta_t \exp(i\mathbf{k}^t \cdot \mathbf{r}) \right] \hat{\mathbf{e}}_{\perp} \quad (1.25b)$$

where $E_{\alpha}^{(j)}$ is the magnitude of either the reflected ($\alpha \rightarrow r$) or transmitted ($\alpha \rightarrow t$) electric field considering the j polarization state; for either case E^{inc} is the magnitude of the incident electric field. Additionally, the incident $\mathbf{B}^{\text{inc}}(\mathbf{r})$ and reflected and transmitted magnetic fields $\mathbf{B}_{\alpha}^{(j)}(\mathbf{r})$ for both polarizations can be determined by substituting Eqs. (1.24) and (1.25) into Faraday-Lenz's Law, while the displacement field and the H field are calculated via the general relations $\mathbf{D} = \varepsilon \mathbf{E}(\mathbf{r})$ and $\mathbf{H}(\mathbf{r}) = \mathbf{B}(\mathbf{r})/\mu_0$ considered for each media. Prior calculation of the total polarization in the film $\mathbf{P}^{\text{film}}(\mathbf{r}_{\parallel})$ it is noteworthy to show that forcing the the proposed electric fields in Eqs. (1.24) and (1.25) to follow the boundary conditions in Eqs. (1.19) yields the continuity of the parallel component of the wave vector $\mathbf{k}_{\parallel} = (\omega/c)n_i \sin \theta_i \hat{\mathbf{e}}_x = (\omega/c)n_t \sin \theta_t \hat{\mathbf{e}}_x$, therefore stating that even with the presence of the thin film, the reflection and Snell's law are valid. Under this consideration, $\mathbf{P}^{\text{film}}(\mathbf{r}_{\parallel})$ is obtained by introducing Eqs. (1.24) and (1.25) into Eq. (1.22) yielding

$$\mathbf{P}_{\parallel}^{(s),\text{film}}(\mathbf{r}_{\parallel}) = \frac{\gamma}{2} (E^{\text{inc}} + E_r^{(s)} + E_t^{(s)}) \exp(\mathbf{k}_{\parallel} \cdot \mathbf{r}) \hat{\mathbf{e}}_x \quad (1.26a)$$

$$P_{\perp}^{(s),\text{film}}(\mathbf{r}_{\parallel}) = 0, \quad (1.26b)$$

for s polarization and

$$\mathbf{P}_{\parallel}^{(p),\text{film}}(\mathbf{r}_{\parallel}) = \frac{\gamma}{2} \left[(E_r^{(p)} - E^{\text{inc}}) \cos \theta_i - E_t^{(p)} \cos \theta_t \right] \exp(\mathbf{k}_{\parallel} \cdot \mathbf{r}) \hat{\mathbf{e}}_x \quad (1.26c)$$

$$P_{\perp}^{(s),\text{film}}(\mathbf{r}_{\parallel}) = \frac{\beta}{2} \left[\varepsilon_i (E^{\text{inc}} + E_r^{(p)}) \sin \theta_i + \varepsilon_t E_t^{(p)} \sin \theta_t \right] \exp(\mathbf{k}_{\parallel} \cdot \mathbf{r}) \quad (1.26d)$$

for p polarization. point dipoles. The electromagnetic behavior of the metasurface is described through the susceptibilities γ and β . Specifically, $\gamma = \rho \alpha_{\parallel}$ relates surface polarization parallel to the interface, while $\beta = \rho \alpha_{\perp} / \varepsilon_s^2$ relates surface polarization perpendicular to it. Here, $\alpha_{\parallel}$ and $\alpha_{\perp}$ denote the particle polarizabilities along their major and minor axes, respectively, and ρ is the particle surface density. These expressions hold under the condition of identical non-interacting nanoparticles, i.e., in the low coverage limit (small ρ). The modified Fresnel coefficients can be formulated in terms of the surface susceptibilities γ and β as [**Bedeaux_Optical, Lazzari_GranFilm**]

$$r_s = \frac{n_i \cos \theta_i - n_t \cos \theta_t + i(\omega/c)\gamma}{n_i \cos \theta_i + n_t \cos \theta_t - i(\omega/c)\gamma}, \quad (1.27)$$

$$r_p = \frac{\kappa_{-}(\omega) - i(\omega/c)\gamma \cos \theta_i \cos \theta_t + i(\omega/c)n_i n_t \varepsilon_i \beta \sin^2 \theta_i}{\kappa_{+}(\omega) - i(\omega/c)\gamma \cos \theta_i \cos \theta_t - i(\omega/c)n_i n_t \varepsilon_i \beta \sin^2 \theta_i}, \quad (1.28)$$

$$\kappa_{\pm}(\omega) = \left(n_t \cos \theta_i \pm n_i \cos \theta_t \right) \left[1 - \left(\frac{\omega}{2c} \right)^2 \varepsilon_i \gamma \beta \sin^2 \theta_i \right] \quad (1.29)$$

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where θ_i and θ_t are the angles of incidence and transmission, respectively, ω is the frequency of the incident light and c is the speed of light. To simplify the notation, the term is introduced. Equations (1.27) and (1.28) are reduced to the well-known Fresnel coefficients when the surface density of Au NI goes to zero (substrate-ambient interface).

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